ETFs Simulation with Temporal Memory: A Positive Fractional–Diffusive Volatility Model

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Research manuscript — 19 October 2026

Abstract

We study a positive conditional-variance model combining finite fractional-integral memory and implicit diffusion on a correlation graph. We establish positivity, fixed-parameter moment stability and exact multi-step variance forecasts for the implemented recursion, without deriving it from an option-pricing equation. A chronological study of the S&P 500 and eleven sector ETFs uses 186 origins during 2023–2026 and horizons of 1, 5 and 21 observations. Adding persistence, asymmetry and multi-horizon validation improves the initial fractional model in all six panel–horizon comparisons, but GJR-GARCH remains stronger. These development results are exploratory. A further protocol fixes eight country ETFs before download and retains the model unchanged. It adds FIGARCH and a fixed 50:50 fractional/GJR forecast ensemble. The ensemble improves pooled QLIKE over the standalone fractional model and FIGARCH in all nine comparisons across the three panels. It exceeds GJR-GARCH in only three, with all paired ranges against GJR including zero. The country panel does not confirm a general diffusion benefit. The contribution is an explicit constrained fractional–diffusive risk model and a reproducible assessment of its properties, forecast combinations and limitations on retrospective adjusted-price data.

Keywords: fractional memory; conditional variance; graph diffusion; sector ETFs; volatility forecasting; chronological validation.

Data and code availability. Code, tests, protocols and aggregate results accompany this draft. The project repository is https://github.com/enia-ardid/mathematical_finance. The accompanying code and manuscript should be identified by their Git commit. Raw prices are retained locally with source metadata and hashes and are not bundled.

Research manuscript for author review; not a peer-reviewed publication.

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1 Introduction

Financial returns provide a difficult setting for directional forecasting: recent prices already constitute a strong reference, while a visually plausible price path can conceal weak predictive information. A more focused question is whether temporal memory and sector interactions help forecast the magnitude of future fluctuations. This paper retains those two ingredients and assigns them to an explicit conditional-variance model.

Fractional filters provide flexible history weighting. Infinite fractional differencing has a well-developed time-series interpretation [1], and FIGARCH introduces hyperbolic persistence into conditional heteroskedasticity [2]. Neither construction implies that a Hurst estimate can be inserted into an arbitrary stochastic recurrence. Likewise, diffusion between economic categories requires a specified interaction operator; an imposed ordering of sector names is not an observed spatial coordinate. Graph operators offer an alternative for irregular domains [3].

Spatial ARCH models already allow conditional heteroskedasticity to depend on neighboring locations [4]. For financial networks, network log-ARCH models combine temporal effects with contemporaneous network interactions [5]. Our specification instead applies a positive graph resolvent to lagged squared innovations with a shared finite fractional kernel. It is a restricted multivariate ARCH model, rather than a new general class of network volatility processes. The specific parameterization and its empirical assessment must therefore be evaluated relative to that existing literature.

Our model applies a normalized finite fractional integral to squared standardized innovations, then smooths this signal through a correlation-graph resolvent. An explicit positive intercept and a feedback constraint control the resulting variance recursion. This formulation differs from both a fractional Black–Scholes equation and FIGARCH. It preserves the project's motivating idea while making its actual stochastic object, initialization and mathematical claims precise.

The contributions are threefold. First, we state and implement a positive fractional–diffusive variance specification, with a graph construction that is equivariant to relabeling assets. Second, we establish positivity, diffusion contraction and fixed-parameter moment properties, and compute multi-step conditional variances without simulation error. Third, we compare the method against historical variance, tuned EWMA, GARCH and two structured ablations using identical chronological forecast origins. These are contributions of formulation and reproducible assessment; the study does not establish priority for the general combination of fractional kernels and graph smoothing.

1.1 Position relative to network volatility models

Network GARCH already uses graph structure to reduce the dimension of multivariate volatility estimation, with stationarity and QMLE results [6]. Threshold network GARCH also incorporates asymmetric shock responses [7]. More recently, generalized network GARCH includes neighboring shock and variance terms alongside covariance forecasts [8]. Neither network dependence nor asymmetry is original to the present specification. Relative to these models, our restriction is a common positive resolvent at every lag and

a normalized fractional temporal profile. Relative to FIGARCH, our finite normalized kernel and subunit feedback define a different recursion; FIGARCH is fitted directly in the additional experiment in Section 8. Published network models are discussed but not fitted, so no comparative performance claim is made against them.

This narrower scope is intentional: an interpretable constrained model, explicit assumptions, exact forecast moments and reproducible comparisons. The mathematical results specialize positive-filter arguments rather than claiming a new general stationarity theorem. The temporal/spatial restriction is made explicit below through its impulse response.

The results support a narrower conclusion than universal predictive superiority. The initial specification improves average losses relative to historical variance and EWMA, but GARCH remains stronger. A persistent asymmetric refinement improves all six pooled comparisons with that first specification and exceeds GARCH at one and five observations in both US panels. GJR-GARCH remains stronger throughout that standalone comparison. The additional external-universe study and fixed forecast combination extend the assessment in Section 8. Diffusion helps the one-session sector average modestly, with uncertainty ranges including zero. Both the improvement and the stronger counterexample are needed to assess the contribution of the fractional and diffusive components.

2 Data, notation and prediction target

2.1 Observed prices and returns

Let $P_{t,j} > 0$ denote the closing price of asset j at retained trading observation t, and define

$$r_{t,j} = \log(P_{t,j}/P_{t-1,j}). \tag{1}$$

The information available at forecast origin o is $\mathcal{F}_o = \sigma(P_0, \ldots, P_o)$. All transformations, graph estimates and parameter choices used to forecast from o depend only on this information.

The market snapshot was downloaded on 17 October 2026 from Yahoo Finance through `yfinance` 0.2.66, with adjusted prices and an exclusive request end date of 17 October 2026. It contains the index `^GSPC` and the eleven ETFs XLB, XLE, XLF, XLI, XLK, XLP, XLRE, XLU, XLV, XLY and XLC. After inception trimming and alignment, common prices run from 19 June 2018 through 16 October 2026. Prices must be positive, finite, uniquely dated and increasing in date; missing interior observations are not filled.

The index and ETF panel are fitted separately. Adjusted ETF returns and an index price-return series are not identical economic objects; we do not interpret their combination as a self-financing index replication. The snapshot is retrospective and may contain adjustment revisions unavailable at historical forecast dates. Chronological code prevents access to future rows, but it does not turn revised data into point-in-time observations.

2.2 Risk target and common mean convention

At each parameter refit, estimate a mean μ_j and a positive scale s_j^2 from the observed fitting window. These estimates are shared by all benchmark models until the next refit. Write

$$u_{t,j} = r_{t,j} - \mu_j, \qquad z_{t,j} = u_{t,j}^2 / s_j^2. \tag{2}$$

The h-observation evaluation target is

$$Y_{o,h,j} = \sum_{\ell=1}^{h} (r_{o+\ell,j} - \mu_j)^2, \qquad h \in \{1, 5, 21\}. \tag{3}$$

This is a daily squared-return proxy, not intraday realized variance. Under the specified constant conditional mean it is conditionally unbiased for the sum of future conditional variances. If the fitted mean differs from the true conditional mean, the target also contains the corresponding squared mean error. We therefore interpret the empirical task conservatively as forecasting centered return second moments.

3 A positive fractional–diffusive model

3.1 Finite fractional-integral memory

For $0 < d \leq 1$, the binomial expansion of $(1 - B)^{-d}$ yields positive coefficients

$$g_0(d) = 1, \qquad g_k(d) = \frac{k - 1 + d}{k}\, g_{k-1}(d) = \frac{\Gamma(k + d)}{\Gamma(d)\Gamma(k + 1)}. \tag{4}$$

With memory length M, normalize these coefficients:

$$p_k(d) = \frac{g_k(d)}{\sum_{v=0}^{M-1} g_v(d)}, \quad m_t = \sum_{k=0}^{M-1} p_k(d) z_{t-1-k}. \tag{5}$$

Thus $p_k > 0$ and $\sum_k p_k = 1$. For large k, $g_k(d)$ behaves as $k^{d-1}/\Gamma(d)$. Within a fixed window, increasing d spreads weight toward older observations; $d = 1$ gives a uniform moving average. The study fixes $M = 252$. It does not infer d from a Hurst estimator.

The normalized operator is a finite-window analogue of a fractional integral. For a smooth signal q, a physical window T and a refining observation step, the corresponding normalized fractional integral is

$$\mathcal{I}_{d,T} q(t) = \frac{d}{T^d} \int_0^T a^{d-1} q(t - a)\, da. \tag{6}$$

This relation motivates the kernel; the empirical model itself is the finite discrete recursion (5). Finite memory and subunit feedback do not establish asymptotic long-range dependence or convergence to a fractional stochastic PDE.

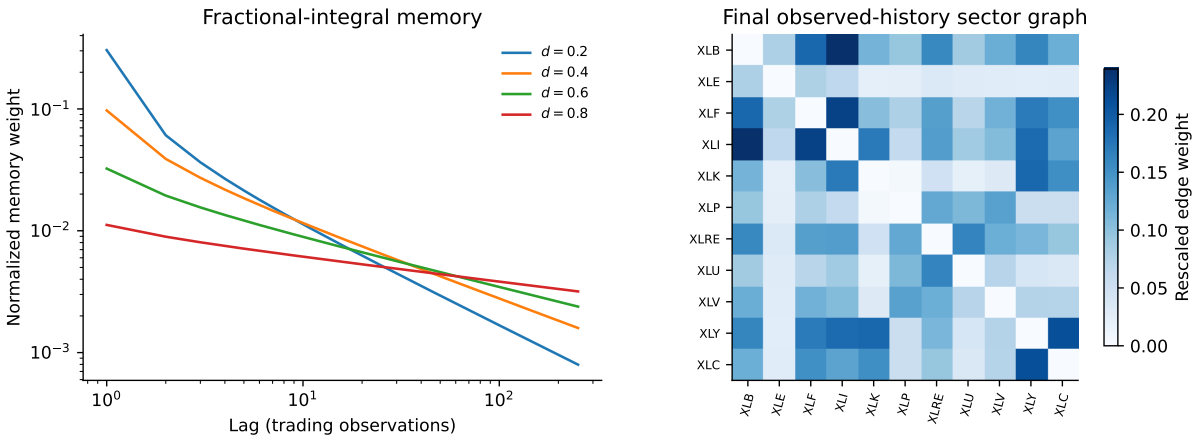


Figure 1: Normalized fractional-integral weights and the sector graph from the last parameter refit in the recorded backtest. The graph uses only the history available at that refit. Weights are shown over the actual 252-observation memory window; an infinite-memory interpretation is not assumed.

3.2 Correlation graph and implicit diffusion

Estimate a Ledoit–Wolf covariance matrix [9] after standardizing the fitting-window returns. Convert it to a correlation matrix R and set

$$W_{ij} = \max(R_{ij}, 0)^2 \quad (i \neq j), \qquad W_{ii} = 0. \tag{7}$$

Let $D = \operatorname{diag}(W\mathbf{1})$ and rescale the graph Laplacian by its mean degree:

$$L = \frac{D - W}{\max(J^{-1} \operatorname{tr} D, 10^{-12})}, \qquad A_\tau = (I + \tau L)^{-1}, \quad \tau \geq 0. \tag{8}$$

For the scalar index, $L = 0$ and $A_\tau = 1$; index results contain no evidence about cross-sector diffusion. For sectors, the update $x = A_\tau m$ is the implicit Euler step $x - m = -\tau L x$ for graph heat diffusion. The diffusion parameter controls smoothing on this estimated network, not a causal contagion rate.

No arbitrary one-dimensional sector ordering is required. Under a permutation matrix Π, the same construction transforms L into $\Pi L \Pi^\mathsf{T}$ and A_τ into $\Pi A_\tau \Pi^\mathsf{T}$. Forecasts are consequently permutation equivariant.

3.3 Conditional variance and return generation

For $0 \leq \eta < 1$, define normalized conditional variance h_t and asset-scale variance v_t by

$$h_t = (1 - \eta)\mathbf{1} + \eta A_\tau m_t, \qquad v_t = s^2 \odot h_t, \tag{9}$$

where $\odot$ denotes elementwise multiplication. Returns obey

$$r_t = \mu + \operatorname{diag}(\sqrt{v_t})\, \varepsilon_t, \quad \mathbb{E}[\varepsilon_t \mid \mathcal{F}_{t-1}] = 0, \quad \mathbb{E}[\varepsilon_{t,j}^2 \mid \mathcal{F}_{t-1}] = 1. \tag{10}$$

For simulations, the implementation uses temporally independent Gaussian innovation vectors with correlation R. Cross-sectional correlation does not invalidate the marginal

moment results below. Gaussian innovations are a simulation assumption, not a claim that market return distributions are Gaussian.

Diffusion acts on standardized variance signals. It preserves their sum, not necessarily the sum of variances in original asset units. This distinction avoids forcing low-volatility and high-volatility sectors to have equal scale. The positive intercept is explicit, and all simulated paths use the same stated recursion rather than a fitted terminal price correction.

To make the connection with multivariate ARCH explicit, let $S = \mathrm{diag}(s_1^2, \ldots, s_J^2)$. In asset units, the lag coefficient matrices are $C_k = \eta p_k S A_\tau S^{-1}$, so that $v_t = (1-\eta)s^2 + \sum_{k=0}^{M-1} C_k(u_{t-1-k} \odot u_{t-1-k})$. All C_k are nonnegative, and $\rho(\sum_k C_k) = \eta$. Fractional weighting and graph diffusion constrain these matrices to an interpretable shared structure.

4 Mathematical properties

Proposition 1 (Positive, nonamplifying graph diffusion)**.** *For a symmetric nonnegative adjacency matrix and $\tau \geq 0$, A_τ is entrywise nonnegative and doubly stochastic. Its eigenvalues lie in $(0, 1]$. For any real vector x,*

$$\min_j x_j \leq (A_\tau x)_i \leq \max_j x_j, \qquad (A_\tau x)^\mathsf{T} L (A_\tau x) \leq x^\mathsf{T} L x. \tag{11}$$

Proof. The Laplacian is symmetric positive semidefinite and $L\mathbf{1} = 0$. Hence $I + \tau L$ is invertible and $A_\tau \mathbf{1} = \mathbf{1}$. For $b \geq 0$, a negative minimum of the solution to $x + \tau L x = b$ would make the corresponding left-hand component negative, a contradiction. Thus $A_\tau \geq 0$. Symmetry gives unit column sums. The componentwise bound follows from convex averaging. Diagonalizing L gives eigenvalues $(1 + \tau\lambda_i)^{-1}$ for A_τ; the energy inequality follows from $\lambda_i/(1 + \tau\lambda_i)^2 \leq \lambda_i$. □

Proposition 2 (Positivity and fixed-parameter moment stability)**.** *Fix $M < \infty$, $s_j^2 > 0$ and $0 \leq \eta < 1$. For nonnegative observed squared innovations, $v_{t,j} \geq (1 - \eta)s_j^2$. If the innovation vectors are iid over time with zero means and unit marginal second moments, the normalized squared-return recursion admits a unique causal stationary solution with finite first moments, and its stationary marginal expectations satisfy $\mathbb{E}z_t = \mathbf{1}$. Consequently, the corresponding stationary returns have variance s_j^2 and finite second moments $\mu_j^2 + s_j^2$.*

Proof. The lower bound follows from nonnegative weights and nonnegative diffusion. If initial component expectations are bounded by $C \geq 1$, then $\mathbb{E}h_{t,j} \leq (1 - \eta) + \eta C \leq C$, and $\mathbb{E}z_{t,j} = \mathbb{E}h_{t,j}$. For two histories driven by the same innovation sequence, put $e_t = \mathbb{E}\|z_t - z'_t\|_1$. Independence of the current innovations from the past, unit marginal second moments and the unit column sums of A_τ give

$$e_t \leq \eta \sum_{k=0}^{M-1} p_k e_{t-1-k}. \tag{12}$$

The largest expected discrepancy contracts by at least η each time an entire block of M initial lags is replaced. Backward iteration from a bounded deterministic history is therefore Cauchy in L^1, yielding a causal stationary limit; the same bound gives uniqueness among finite-first-moment solutions. In stationarity, $b = \mathbb{E}z_t$ solves $b = (1 - \eta)\mathbf{1} + \eta A_\tau b$. Since $\rho(\eta A_\tau) < 1$, its unique solution is $b = \mathbf{1}$. □

These results concern fixed parameters. Rolling estimates need not form a stationary process, and finite second moments do not imply finite fourth moments, asymptotic normality of loss averages, or guaranteed forecast accuracy. The finite-window contraction also explains why a power-shaped kernel should not be advertised as a proof of infinite-horizon long memory.

5 Forecasting and estimation

5.1 Exact multi-step marginal moments

At origin o, let $\bar{z}_{o+\ell|o}$ equal observed z for nonfuture lags, and recursively replace future squared innovations by their conditional expectations:

$$\bar{h}_{o+\ell|o} = (1-\eta)\mathbf{1} + \eta A_\tau \sum_{k=0}^{M-1} p_k \bar{z}_{o+\ell-1-k|o}, \qquad \bar{z}_{o+\ell|o} = \bar{h}_{o+\ell|o}. \tag{13}$$

Linearity and unit innovation second moments make this recursion exact under the fitted variance model. The aggregate forecast is

$$\widehat{V}_{o,h} = s^2 \odot \sum_{\ell=1}^{h} \bar{h}_{o+\ell|o}. \tag{14}$$

With the model's conditional mean, this also gives the marginal conditional variance of cumulative future log-returns because the centered increments are martingale differences. It does not provide an exact Gaussian predictive law for their sum: future variances are themselves random. The study evaluates variance forecasts directly and does not exponentiate an assumed Gaussian multi-step law into a purported exact expected price.

5.2 Chronological selection

Forecast origins occur every five retained observations on a grid beginning at price position 252; scoring begins in 2023. At the first scored origin and whenever at least 21 observations have elapsed since the last refit, parameters are updated. This gives a 25-observation refit spacing on the five-observation grid, with 38 refits per panel in the reported experiment.

At a refit origin, reserve the final 252 observed returns for validation. Means, scales and graph structure used to compute validation predictions are estimated strictly before that block, using at most 1,008 returns. As validation advances, already observed squared returns may update the memory state. Their future counterparts are never accessed. Select the minimum pooled one-step QLIKE over

$$d \in \{0.2, 0.4, 0.6, 0.8\}, \qquad \eta \in \{0.8, 0.95, 0.99\}, \tag{15}$$

$$\tau \in \{0, 0.25, 1, 4\}. \tag{16}$$

The index fixes $\tau = 0$. Stable candidate order resolves exact ties. After selection, refit means, scales and the graph on the last 1,008 available returns, including validation observations, and forecast subsequent targets. This is a chronological selection design, not an untouched final test: the historical period was available during earlier research iterations.

The same selected parameters serve all three horizons. Consequently, favorable one-step validation does not guarantee optimal 21-step predictions. All methods are scored at the same origins, excluding end blocks without complete targets for every horizon.

5.3 Benchmarks and ablations

1. **Historical variance:** the current fitting-window sample variance, held constant over each forecast horizon.
2. **EWMA:** squared-innovation averaging with decay selected from $\{0.9, 0.94, 0.97, 0.99\}$ on the same validation block. Its future conditional variance remains constant under its own update equation.
3. **GARCH(1,1):** independent asset models estimated by Gaussian quasi-maximum likelihood [10]. Positivity is imposed and $\alpha+\beta < 0.999$. Two fixed initializations are used; all 456 selected asset-level fits report convergence in this run. No benchmark fit is replaced silently by a weaker model.
4. **Fractional only:** the proposed kernel and feedback with $\tau = 0$, with its own validation-selected d and η.
5. **Exponential plus graph:** replace the fractional average by EWMA while retaining positive feedback and graph diffusion. Its decay, feedback and diffusion grids have the same candidate count as the fractional graph model.

The last two comparisons distinguish the contribution of graph smoothing from the shape of the memory kernel. They are stronger checks than comparing only against a constant price or historical variance.

6 Evaluation of the initial specification

6.1 Loss and dependence

The primary score is a normalized QLIKE loss,

$$\mathcal{L}_{o,h,j} = \log\left(\frac{\widehat{V}_{o,h,j}}{hs_j^2}\right) + \frac{Y_{o,h,j}}{\widehat{V}_{o,h,j}}. \tag{17}$$

The normalization is known at the origin and common to all methods. It changes the level, not paired comparisons. QLIKE is appropriate for conditional second-moment forecasting with noisy, conditionally unbiased proxies [11]; that robustness does not remove sampling noise or mean-model error. Scores can be negative, and ratios of score levels are not reported as percentage improvements. A normalized squared-error diagnostic is also retained in the machine-readable results.

There are 186 common origins from 9 January 2023 to 17 September 2026, with targets through 16 October 2026. The sector panel contributes 2,046 asset–origin pairs per horizon, which must not be treated as independent replications. Twenty-one-observation targets overlap. Paired circular block-bootstrap ranges use 2,000 replicates, seed 1729, and blocks of five or ten origins. Entire sector vectors are resampled together. These are nominal descriptive ranges; they do not correct for all model selection, multiple comparisons, nonstationarity or previous researcher exposure.

panel	index			sectors		
horizon	1	5	21	1	5	21
Historical	0.8368	0.6695	0.6737	0.8167	0.7187	0.7222
EWMA	0.8397	0.4576	0.5602	0.7195	0.5914	0.6251
GARCH	0.6004	0.3640	0.4714	0.6297	0.5499	0.6021
Fractional only	0.6514	0.4223	0.5467	0.6549	0.5638	0.6175
Exponential + graph	0.6690	0.3767	0.4841	0.6524	0.5542	0.5953
Fractional + graph	0.6514	0.4223	0.5467	0.6405	0.5627	0.6196

Table 1: Mean normalized QLIKE at 1, 5 and 21 observations. Lower is better. All methods use identical origins and the same mean convention. The index has no cross-sector diffusion, so its two fractional rows coincide.

6.2 Pooled comparison

Table 1 shows lower mean loss for fractional diffusion than tuned EWMA at every panel–horizon combination. The one-session differences are -0.18834 for the index and -0.07899 for sectors. Nevertheless, GARCH has lower mean QLIKE in all six cases. For sectors, the proposed method's excess losses relative to GARCH are 0.01084, 0.01287 and 0.01752 across the three horizons. These findings support a useful alternative representation, not dominance over standard conditional-variance models.

Graph diffusion changes the one-session sector score from 0.65489 to 0.64053. Its gain is much smaller at five observations, and the no-diffusion model is slightly better at 21 observations. Nonzero diffusion is selected at 31 of 38 sector refits, but validation selection is not itself evidence of outer-sample predictive benefit. The exponential graph model outperforms fractional diffusion at five and 21 observations in both panels; the fractional kernel has the lower one-session mean in both panels. Memory shape matters, but no shape dominates all horizons.

With ten-origin blocks, the nominal 95% range for the one-session sector difference from EWMA is $[-0.26077, 0.03470]$. Relative to no diffusion, it is $[-0.03672, 0.00145]$. Both include zero. The difference from GARCH is 0.01084, with range $[-0.03822, 0.04668]$. We therefore do not infer a reliable edge over these dynamic references. Full ranges for both block lengths are distributed with the generated tables.

6.3 Variation over time

Average improvements are not uniform over calendar years. For the index, the proposed method loses to EWMA at all horizons in 2023, wins at all horizons in 2024 and 2026, and has mixed results in 2025. For sectors, it loses at all horizons in 2023 and improves on EWMA in the later years. The 2025 index one-session difference of -0.5367 materially influences the pooled average; it should not be interpreted as a stable gain in every regime.

Figure 3 illustrates the response to changing fluctuation levels. The plot is descriptive and supplements the numerical losses; visual agreement alone is not a forecast test. Annual scores and losses for every asset are recoverable from the origin-level local outputs, while aggregate annual tables are included in the manuscript bundle.

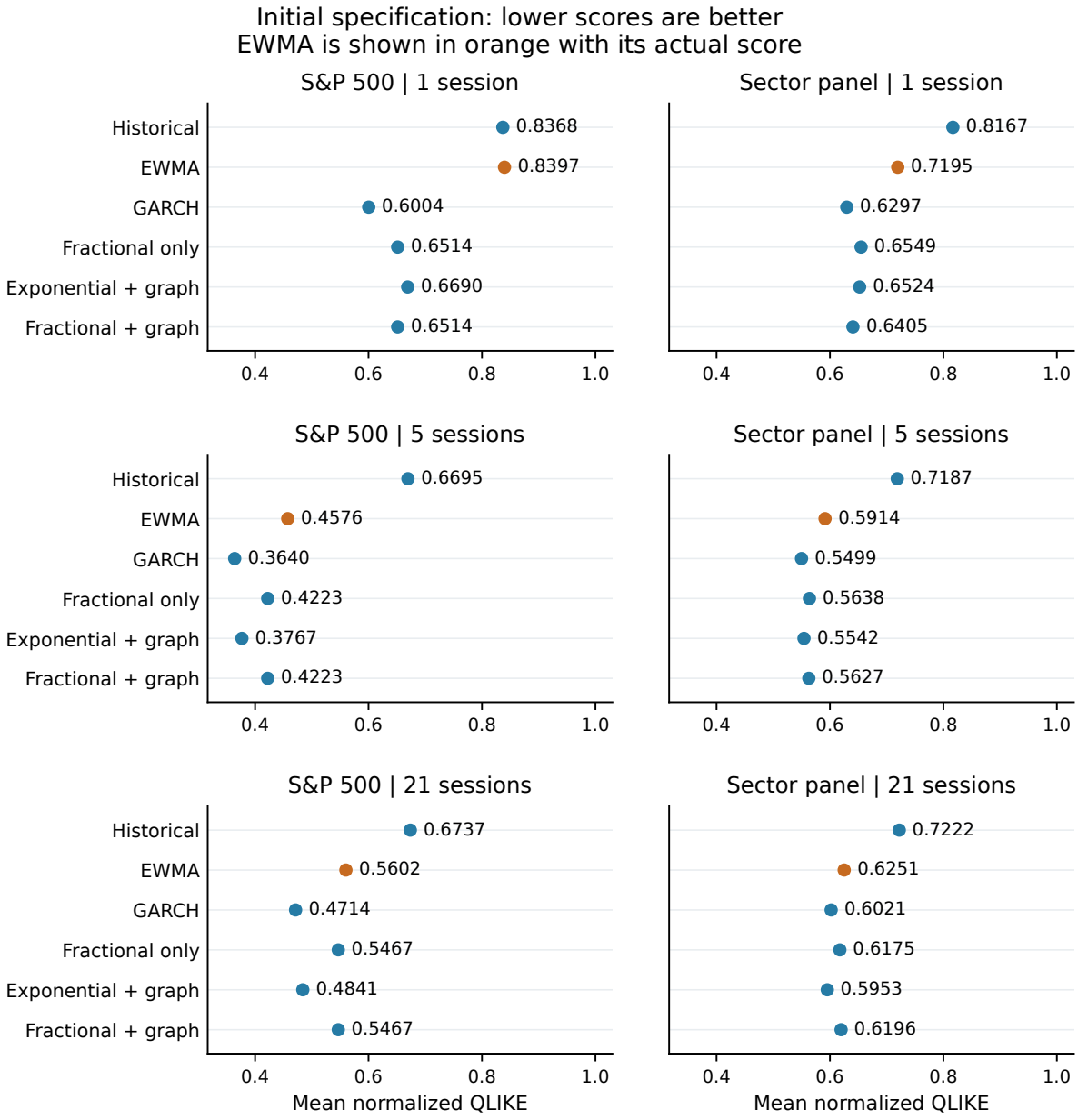


Figure 2: Mean normalized QLIKE for the initial specification and references. Lower values indicate smaller loss. Each point and its numerical label show the actual score, including EWMA in orange. The six panels share the same horizontal scale and reproduce Table 1; all methods use identical origins within each panel and horizon.

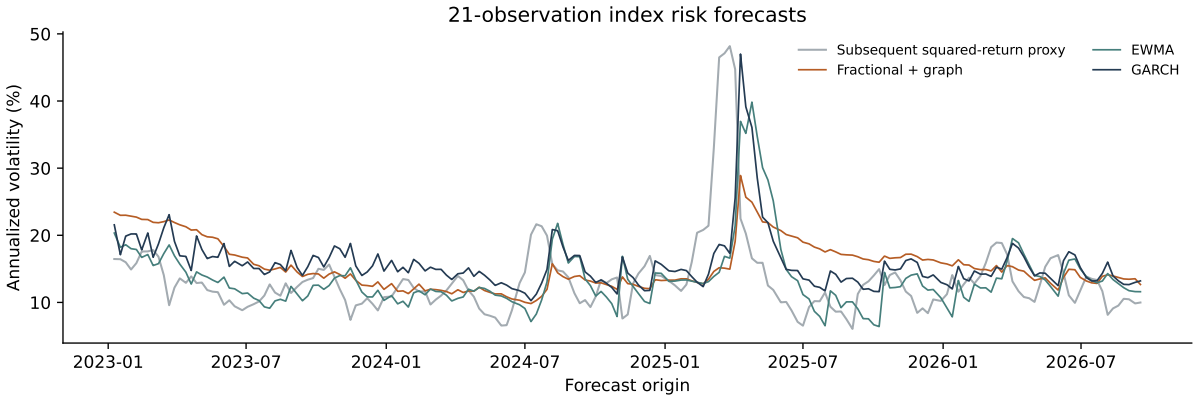


Figure 3: Index risk forecasts over 21 subsequent observations, displayed in annualized volatility units. The gray series is the subsequently observed squared-return proxy, plotted at its forecast origin for comparison; it was not available then. The proxy is noisy and is not the latent conditional variance.

7 Persistence and asymmetric innovations

7.1 Specification and moment properties

The initial model responds identically to positive and negative innovations and selects parameters using only one-step losses. We retain its fractional kernel and graph diffusion while allowing a persistent variance state and asymmetric shock contributions. Define

$$x_{t,j} = z_{t,j}\frac{1+\delta\mathbf{1}\{u_{t,j}<0\}}{1+\delta/2}, \tag{18}$$

$$h_t = (1-b)(1-\eta)\mathbf{1} + bh_{t-1} + (1-b)\eta A_\tau \sum_{k=0}^{M-1} p_k x_{t-1-k}, \tag{19}$$

with $v_t = s^2 \odot h_t$, $0 \le b < 1$ and $\delta \ge 0$. The parameter b smooths the variance response. With $\delta = 1$, a negative innovation has twice the contribution of a positive innovation of the same magnitude.

For the asymmetric recursion, assume conditionally symmetric innovations with unit second moments. This is stronger than a zero-mean, unit-variance assumption: symmetry gives $\mathbb{E}[\varepsilon_{t,j}^2\mathbf{1}\{\varepsilon_{t,j}<0\}] = 1/2$ and therefore $\mathbb{E}[x_t \mid \mathcal{F}_{t-1}] = h_t$. Gaussian simulation innovations satisfy the assumption; it is not asserted as an empirical property of market shocks. Future expected variances are obtained by the same substitution as in Section 5.1, retaining the additional state h_{t-1}.

Unit initial variance and unit presample transformed squares give the floor $v_{t,j} \ge (1-\eta)s_j^2$ by induction. For fixed parameters and iid symmetric innovations, the shared-noise argument for Proposition 2 now gives, for discrepancies between normalized variance processes,

$$e_t \le be_{t-1} + (1-b)\eta \sum_{k=0}^{M-1} p_k e_{t-1-k}, \qquad b + (1-b)\eta < 1. \tag{20}$$

The maximum expected discrepancy over the finite lag state contracts over successive M-observation blocks. The same backward-iteration argument yields a unique causal finite-

first-moment stationary variance process. Its mean is $\mathbf{1}$, so return variance is s_j^2. The recursive state adds exponentially decaying persistence; it does not establish asymptotic long-range dependence.

7.2 Effective memory and graph modes

The persistence state has an exact filter interpretation. For fixed parameters, unroll (19) after removing its initialization transient to obtain

$$h_t = (1-\eta)\mathbf{1} + \eta A_\tau \sum_{j=1}^{\infty} \widetilde{p}_j x_{t-j}, \tag{21}$$

$$\widetilde{p}_j = (1-b) \sum_{k=0}^{\min(M-1,j-1)} p_k b^{j-1-k}. \tag{22}$$

Indeed, repeated substitution expands the state into a geometric sum; grouping terms with the same innovation lag yields (22). Since all terms are nonnegative, summing in either order gives $\sum_{j\geq 1} \widetilde{p}_j = (1-b)\sum_k p_k \sum_{\ell\geq 0} b^\ell = 1$. For $0 < b < 1$ and $j \geq M$, $\widetilde{p}_{j+1} = b\widetilde{p}_j$. At $b = 0$ the weights reduce to p_{j-1} for $1 \leq j \leq M$, with zero tail. Thus the refinement geometrically smooths the finite fractional kernel; its infinite tail, when present, is exponential rather than hyperbolic.

With $L = U\Lambda U^\mathsf{T}$, the graph-mode coefficient at lag j is $\eta\widetilde{p}_j/(1+\tau\lambda_\ell)$. Temporal weighting and spatial smoothing are separable: every lag shares the same graph attenuation. This clarifies both interpretability and a restriction relative to more general network lag matrices. A numerical impulse-response test verifies the identity against the implemented filtering recursion.

7.3 Recorded follow-up protocol

This refinement was designed after inspecting the initial study. The separate `paper/refinement_protocol.json` records its rules before its candidates were evaluated, but the historical targets are reused and no independent confirmation is claimed. At each refit, use the same 252-observation validation block and fit nuisance parameters strictly before it. Retain only validation origins whose complete 21-observation targets finish by the refit date. Minimize the equal-weight mean of normalized QLIKE at horizons 1, 5 and 21, using all eligible daily origins. A single selected specification serves all outer horizons.

Retain the earlier grids for d, η and τ, and add $b \in \{0, 0.5, 0.9\}$ and $\delta \in \{0, 1\}$. There are 288 candidates for sectors and 72 for the scalar index per kernel family. A symmetric variant fixes $\delta = 0$; a graph ablation fixes $\tau = 0$; a kernel ablation uses normalized exponential weights truncated to the same 252 observations, with the earlier four decay values. Each ablation has its own validation selection. The larger candidate set increases selection risk despite chronological ordering.

An additional GJR-GARCH reference permits asymmetric shock responses [12]. Its normalized variance satisfies $h_t = \omega + (\alpha + \gamma\mathbf{1}\{u_{t-1} < 0\})z_{t-1} + \beta h_{t-1}$. Independent asset fits use Gaussian QMLE, two fixed starts, nonnegative coefficients and $\alpha + \beta + \gamma/2 < 0.999$.

panel	index			sectors		
horizon	1	5	21	1	5	21
Original fractional	0.6514	0.4223	0.5467	0.6405	0.5627	0.6196
Persistent symmetric	0.6143	0.3815	0.5162	0.6447	0.5582	0.6116
Persistent asymmetric	0.5819	0.3574	0.4977	0.6242	0.5478	0.6043
Without graph	0.5819	0.3574	0.4977	0.6317	0.5474	0.6046
Exponential kernel	0.6223	0.3516	0.4785	0.6656	0.5588	0.5980
EWMA	0.8397	0.4576	0.5602	0.7195	0.5914	0.6251
GARCH	0.6004	0.3640	0.4714	0.6297	0.5499	0.6021
GJR-GARCH	0.5251	0.3128	0.4360	0.6053	0.5401	0.5931

Table 2: Mean normalized QLIKE for the refinement, its ablations and references. "Original fractional" denotes the initial variance specification. All rows use the same 186 origins. Lower is better.

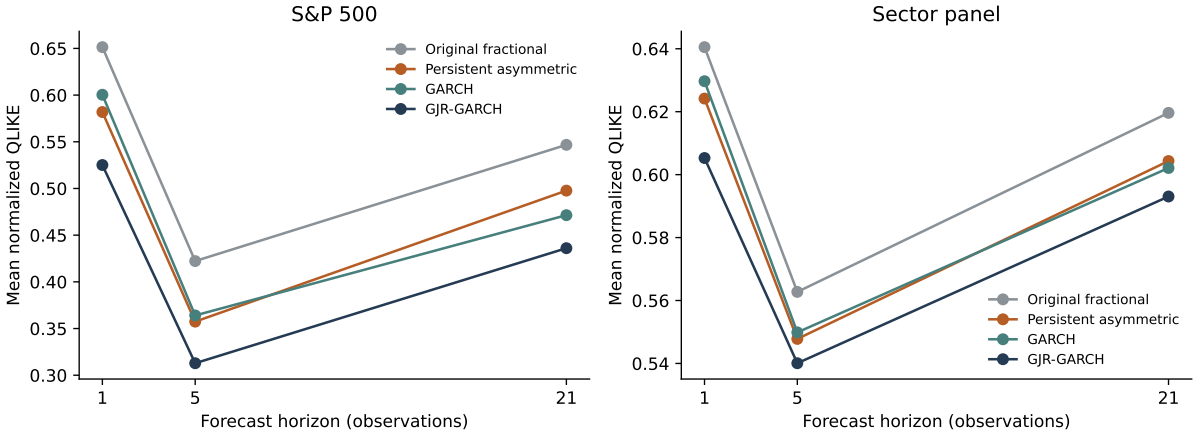


Figure 4: Pooled losses before and after the refinement, alongside GARCH and GJR-GARCH. The asymmetric reference remains stronger in every displayed case.

All 456 selected fits report convergence. Multi-step moments also require conditional symmetry. Earlier reference forecasts are recomputed; all 40,176 original forecast records and loss values match exactly, so comparisons use unchanged targets and scales.

7.4 Follow-up results

Table 2 shows lower pooled loss for persistent asymmetric fractional diffusion than the initial model in all six comparisons. Index scores change from $(0.65138, 0.42225, 0.54672)$ to $(0.58187, 0.35738, 0.49770)$, and sector scores from $(0.64053, 0.56274, 0.61963)$ to $(0.62417, 0.54779, 0.60432)$. The refinement also has lower loss than GARCH at one and five observations in both panels. GARCH remains better at 21 observations, while GJR-GARCH is better at every panel–horizon combination. These findings establish improvement within the studied fractional family without establishing dominance over asymmetric volatility references.

Using ten-origin blocks, nominal 95% ranges for the change relative to the initial fractional model exclude zero in five of six comparisons; the exception is one-session sector forecast-

ing. For the index, the one-session change is -0.06952 with range $[-0.14531, -0.00876]$. All ranges comparing the refinement with GARCH include zero. These descriptive ranges do not adjust for the repeated research on this snapshot and should not be read as independent confirmation.

Asymmetry is selected at all 38 index refits and 35 sector refits; nonzero variance persistence is selected at every refit. The graph is selected with nonzero strength at 18 sector refits. Relative to its graph ablation, the refinement improves the one-session sector mean by 0.00752, slightly worsens the five-session mean and changes the 21-session mean little. The exponential ablation is stronger for index horizons 5 and 21 and sector horizon 21. Thus neither graph smoothing nor the fractional kernel uniformly dominates its alternative. Annual improvements also vary: relative to the first specification, index loss rises at 21 sessions in 2024, and sector loss rises at 5/21 sessions in 2024 and at one session in 2025. All annual results remain in the public tables.

8 External-universe replication and a fixed forecast combination

8.1 A specification fixed before the new download

The previous two studies reuse US history inspected during development. To assess transfer to another universe, a third protocol fixed eight country ETFs before downloading their prices: EWA (Australia), EWC (Canada), EWG (Germany), EWH (Hong Kong), EWJ (Japan), EWL (Switzerland), EWQ (France) and EWU (United Kingdom). These are long-established US-listed iShares equity funds, selected for geographical spread. The request covers 1 January 2018 through 16 October 2026, using the same adjusted-price convention. All eight assets are required; missing prices are not filled and assets are not removed after seeing results.

The unchanged fractional implementation, parameter grids, validation windows, refit schedule and three horizons are frozen in `paper/external_protocol.json`, with source hashes. No external loss is used to revise that specification. This is a locally timestamped protocol, not an independently registered preregistration. The country panel was not previously evaluated in this project, but it shares global shocks, US trading hours and dollar denomination with the development assets. It is therefore an external-universe replication on retrospective data, not a statistically independent sample or a prospective experiment.

A computational amendment is recorded in the accompanying protocol files. The first attempt reached the FIGARCH forecast and failed because a strided panel column was not C-contiguous. The correction copies input arrays to the required layout. It changes no model or evaluation rule; no FIGARCH comparison results existed at that point. The original protocol and failed-run record are retained.

8.2 Additional forecaster and fractional benchmark

The additional forecaster is deliberately fixed:

$$\widehat{V}^{\mathrm{ens}}_{o,h,j} = \tfrac{1}{2}\widehat{V}^{\mathrm{frac}}_{o,h,j} + \tfrac{1}{2}\widehat{V}^{\mathrm{GJR}}_{o,h,j}. \tag{23}$$

Here the fractional component is the persistent asymmetric model, including its selected graph diffusion. The coefficient is always one half, with no fitted or retrospectively selected combination weights. Positivity is preserved. The ensemble is an optional forecast combination, not a new generative fractional recurrence, and does not replace the standalone model. Its QLIKE is computed from the combined variance; component losses are not averaged. Forecast combinations are not themselves a novelty claim.

FIGARCH(1,d,1) [2] is added as an estimated benchmark using `arch` 7.2.0, Gaussian QMLE and the library's 1,000-lag ARCH approximation. Each asset is centered and standardized using the same origin-fitted mean and scale as the other methods. Fit on the preceding 1,008 returns at the same refit dates, with default starting estimates and SLSQP limited to 1,000 iterations. Between refits, keep parameters and the fitting backcast fixed while updating on observed returns. Analytic forecasts are transformed back to asset units and summed over the same horizons. Nonfinite results abort the experiment; finite nonconverged fits are reported rather than silently replaced. FIGARCH's approximation and initialization differ from our normalized 252-lag kernel and are recorded explicitly.

The same ensemble and FIGARCH reference are appended to the recorded US forecasts after checking their code and input hashes. Those comparisons remain exploratory. Every table uses identical targets within its panel; pooled, annual and asset-level losses are retained. Paired uncertainty ranges use the same whole-panel circular bootstrap and the same nominal interpretation as the earlier study.

8.3 Results

The international panel contains 2,210 prices per asset. Evaluation has 186 origins from 6 January 2023 to 16 September 2026, with complete targets through 15 October 2026 and 38 parameter refits. Its five-observation grid has a different calendar phase from the US grid; comparisons are paired within panels. All 760 FIGARCH asset fits across the three panels converge, as do all 304 international GARCH and 304 GJR-GARCH fits.

The unchanged fractional model obtains international QLIKE $(0.52898, 0.66359, 0.67147)$ at horizons $(1, 5, 21)$, below both EWMA and FIGARCH at every horizon. However, GARCH and GJR-GARCH are stronger at all three. The no-graph ablation has lower loss throughout, while the exponential kernel is better at horizons 5 and 21. Thus the second universe does not support a general benefit from diffusion or a uniform fractional-kernel advantage.

The fixed ensemble lowers those fractional scores to $(0.51307, 0.65094, 0.66134)$, improving all three external pooled comparisons. Its asset-level loss is lower in 22 of 24 country–horizon comparisons, but annual pooled loss worsens relative to the standalone fractional model at horizons 5 and 21 in 2024. Across the US and international panels, the ensemble improves on both the standalone fractional model and FIGARCH in all nine panel–horizon comparisons. These are loss differences, not percentages of an arbitrarily normalized score.

horizon	1	5	21
EWMA	0.5321	0.6700	0.6823
GARCH	0.5287	0.6587	0.6672
GJR-GARCH	0.5083	0.6496	0.6623
FIGARCH	0.5438	0.6671	0.6730
Fractional	0.5290	0.6636	0.6715
Without graph	0.5234	0.6633	0.6701
Exponential kernel	0.5290	0.6526	0.6653
Fixed 50:50 ensemble	0.5131	0.6509	0.6613

Table 3: Mean normalized QLIKE on the eight pre-specified country ETFs. The fractional row is the unchanged persistent asymmetric model. The fixed ensemble averages its variance forecasts with GJR-GARCH. Lower is better.

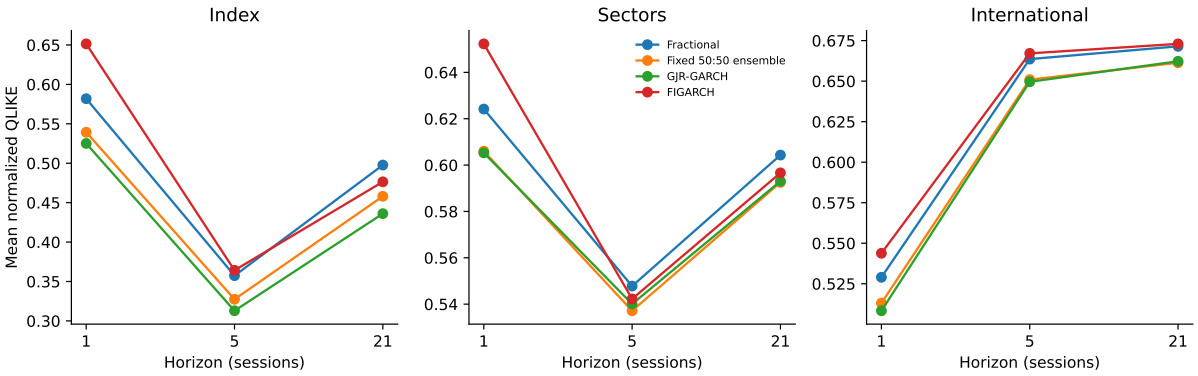


Figure 5: Standalone fractional forecasts, the fixed ensemble, GJR-GARCH and FIGARCH. The index and sector panels reuse the development snapshot; the international panel follows the separately recorded external-universe protocol.

Relative to GJR-GARCH, the ensemble wins only for US sectors at 5 and 21 observations and for the international panel at 21 observations. The international 21-session difference is -0.00092, with ten-origin-block nominal 95% range $[-0.01174, 0.01110]$. All nine ensemble-versus-GJR ranges include zero. For international one-session forecasts, the change from the standalone fractional model is -0.01590; the interval is $[-0.03100, -0.00219]$.

Compared with FIGARCH, the difference is -0.03073. Its interval runs from -0.05536 to -0.01145. These descriptive intervals support reporting the observed improvement without claiming reliable superiority over GJR-GARCH. Full annual, asset-level and paired comparisons remain in the public outputs.

8.4 Observed evolution and rolling predictions

Aggregate losses summarize accuracy but do not show how forecasts respond over time. Figures 6 and 7 compare the recorded predictions with the subsequently observed squared-return targets over the full evaluation period. For readability, both are expressed in

annualized volatility units:

$$\widehat{\sigma}^{\mathrm{ann}}_{o,h,j} = 100\sqrt{\frac{252}{h}\widehat{V}_{o,h,j}}, \qquad \sigma^{\mathrm{proxy}}_{o,h,j} = 100\sqrt{\frac{252}{h}Y_{o,h,j}}. \tag{24}$$

These are square-root transformations of variance forecasts and noisy targets, not estimates of an observed latent volatility path. In particular, the square root of expected variance is not generally expected volatility. At one session, the observed series is a scaled absolute centered return and is especially noisy.

The new time-series panels use the *end of the target window* on the horizontal axis. A prediction plotted at $o+h$ was issued at o, using only the data available then. Curves join the retained origins, spaced five observed sessions apart; they are not a new forecast issued every day. No further fitting, time smoothing, clipping of peaks or favorable subperiod selection is applied. The 21-session target windows overlap.

For the multi-asset displays, average $\widehat{V}_{o,h,j}$ and $Y_{o,h,j}$ over all assets before applying (24). Averaging volatilities directly would give a different quantity. The panel curves can also mask opposing asset-level errors. Appendix D therefore retains the complete set of 21-session paths for every sector and country ETF. These visual diagnostics complement the common-target losses and paired comparisons; visual tracking alone does not establish predictive superiority.

The index curves also reveal limitations hidden by pooled scores. The abrupt 2025 increase is incompletely anticipated, and the persistent forecasts can remain elevated after the observed proxy falls. Smoother predictions should not be interpreted as pointwise reproduction of the realized fluctuations.

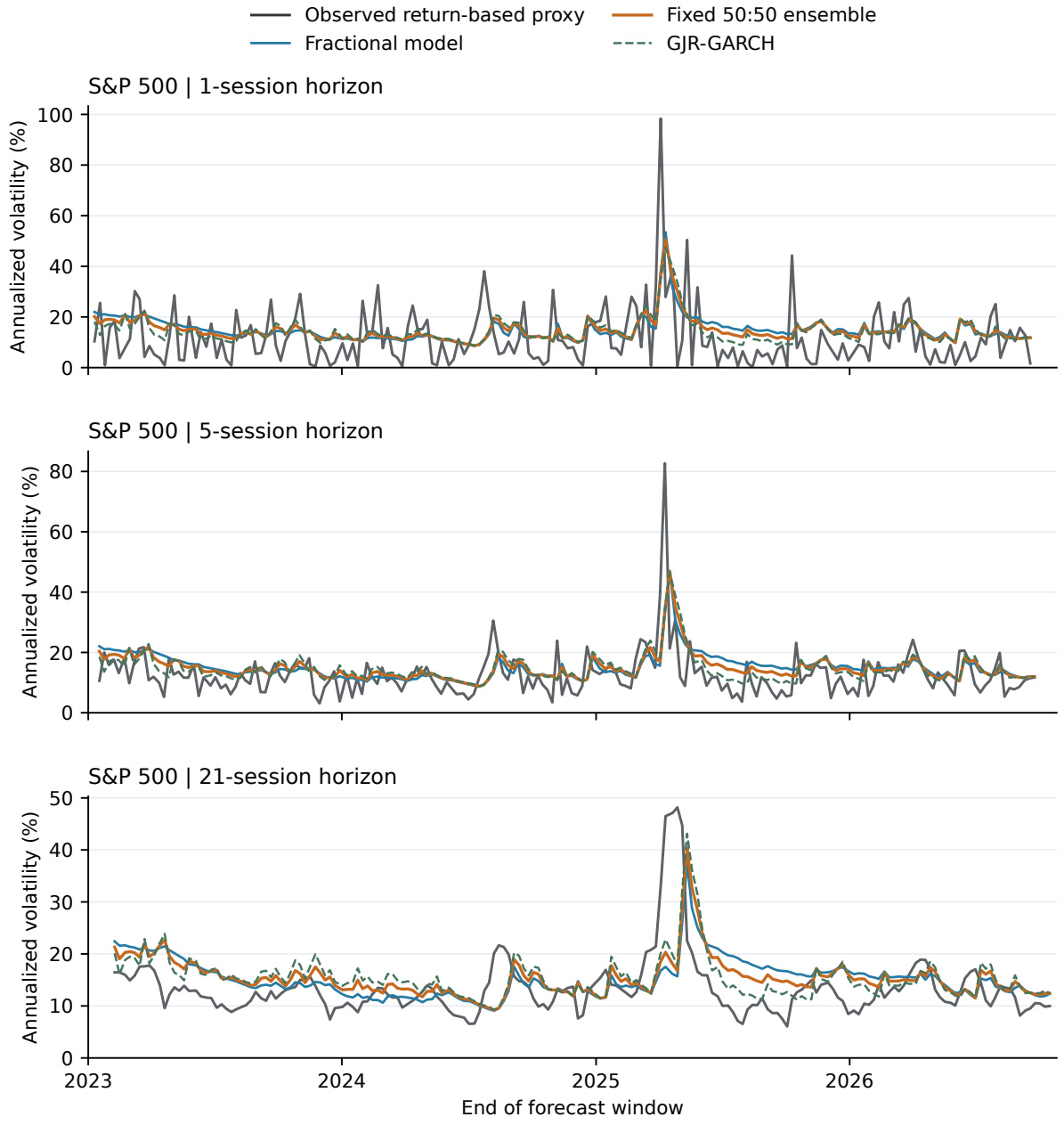


Figure 6: S&P 500 observed proxy and rolling predictions at 1, 5 and 21 sessions. Black denotes the subsequent return-based observation; blue is the persistent fractional model, orange the fixed ensemble, and dashed green GJR-GARCH. Each forecast was issued the stated number of observations before its plotted window-end date. Vertical scales differ by horizon to retain the full peaks.

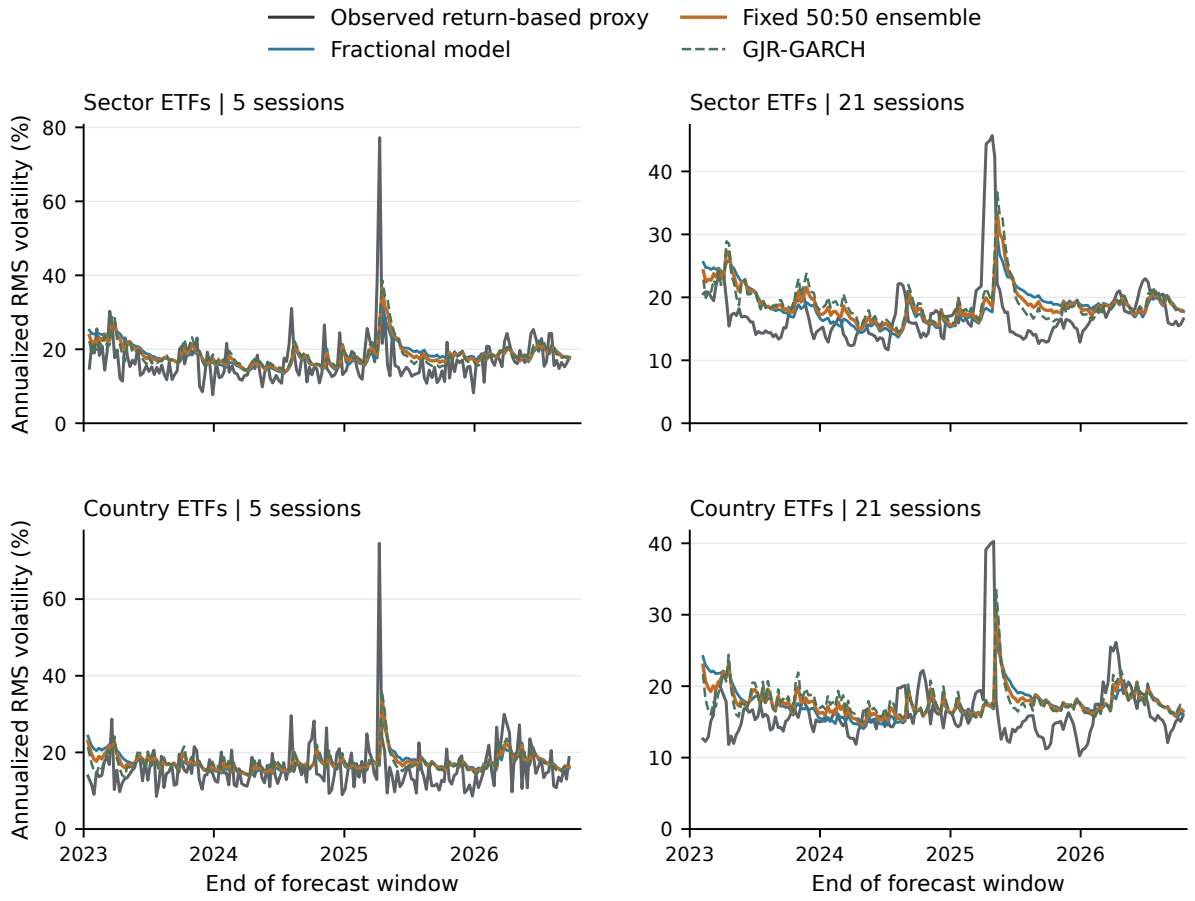


Figure 7: Observed and predicted risk evolution for all eleven sector ETFs and all eight country ETFs, at horizons of 5 and 21 sessions. Marginal variances are averaged across the complete panel before taking the square root. These are annualized root-mean-square marginal volatilities, not portfolio volatility; cross-asset covariances are not included in this display.

9 Interpretation, limitations and conclusions

The revised model preserves temporal fractional weighting and sector diffusion while assigning both to a well-defined positive variance process. The graph step is stable for every nonnegative diffusion strength, and the fixed-parameter recursion has a finite-second-moment return interpretation. These are meaningful improvements in mathematical specification. They do not repair the original price-PDE derivation by assertion: the present model is a new variance specification, and the earlier simulator is retained separately for continuity.

The empirical conclusions are limited by four features of the study. First, the development and international panels use retrospective adjusted prices and share market shocks; neither is a prospective evaluation. Second, daily squared returns are noisy risk proxies. Third, the finite memory window, parameter grids and constant conditional mean restrict the model. Fourth, GJR-GARCH and several kernel-ablation comparisons remain stronger, while descriptive uncertainty ranges often include zero. FIGARCH is now fitted directly, but published network log-ARCH and network GARCH specifications are not; no advantage over those families is established.

Further work should retain point-in-time inputs and evaluate genuinely future targets, with richer volatility proxies and direct network-model benchmarks. Graph estimation uncertainty, richer conditional innovation laws and horizon-specific calibration remain open extensions. Claiming a causal contagion mechanism, a universally superior forecaster, profitable trading, or an infinite-memory limit would exceed the present evidence.

The supported conclusion is that a positive fractional-integral filter with implicit graph diffusion is a coherent and reproducible model for ETF risk experiments. Persistence and normalized asymmetric responses improve the initial fractional specification in all six US pooled comparisons. A fixed combination with GJR further improves the standalone model and FIGARCH in all nine comparisons across three panels. Its small wins against GJR do not establish superiority, and the country-ETF replication does not confirm a general graph benefit. The model is a coherent risk-forecasting component with documented limitations; it does not establish directional price accuracy.

A Implementation and reproduction

The initial implementation is in `fractional_diffusion.py`; the benchmark and asset generator are in `run_paper.py`. The follow-up uses `refined_fractional.py` and `run_refinement.py`. External replication uses `external_validation.py`, `download_external.py` and `run_external.py`. Existing return simulators and forecast heads remain separate. The recorded protocol is `paper/protocol.json`. Input hashes, candidate scores, graph estimates, selected parameters, optimizer statuses and each origin's forecasts are saved. Raw prices and detailed forecast records remain in ignored local artifacts.

```
python run_paper.py --synthetic --output artifacts/paper-synthetic
python run_paper.py --data-dir artifacts/yahoo-through-2026-10-16 \
  --output artifacts/paper-study --publish-assets
python run_refinement.py --synthetic --output artifacts/refinement-synthetic
```

```
python run_refinement.py --data-dir artifacts/yahoo-through-2026-10-16 \
  --output artifacts/refinement-study --publish-assets
python -m unittest discover -s tests -v
cd paper
tectonic --reruns 3 main.tex
```

The synthetic command uses 1,300 correlated Gaussian price observations with no imposed time-varying volatility or fractional mechanism. It is an offline execution example, not empirical evidence for the proposed model. The market snapshot download command and adjustment convention are documented in the data guide. External-study commands and source-package instructions are in `paper/README.md`. The build uses ordinary LaTeX article typography, numbered equations, a contents page and running headers, preserving the presentation of the original project report.

The combined suite contains 53 passing tests on Python 3.12 under macOS. Checks cover fractional weights, graph positivity and contraction, asset relabeling, GARCH/GJR filtering, exact moments against Monte Carlo, asymmetric normalization and validation forecasts against individually truncated histories. Future-price perturbations verify that changing unobserved observations cannot alter earlier variances, selected parameters or graphs. Additional checks cover FIGARCH library equivalence on strided panel inputs, matched ensemble scoring and the effective-memory impulse response. Numerical checks support implementation fidelity; mathematical claims rest on their stated assumptions and proofs.

The presentation script `render_prediction_figures.py` regenerates the direct-score comparison and observed-versus-forecast panels from saved records, without refitting models. It is run after the numerical exporters and before compiling this manuscript; input and output hashes accompany the figures.

B Annual scores

panel	index			sectors		
horizon	1	5	21	1	5	21
year						
2023	0.0021	-0.0541	-0.0447	0.2066	0.1643	0.1675
2024	0.8771	0.2769	0.3731	0.3427	0.3209	0.3531
2025	1.2053	0.9798	1.3312	1.1208	0.9419	1.1219
2026	0.4703	0.5115	0.5196	0.9897	0.9254	0.9201

Table 4: Annual mean normalized QLIKE for the fractional graph model. Levels reflect both realized fluctuations and the origin's reference scale. Negative scores are permissible; comparisons should use common-origin loss differences.

C Relation to the original model

The original project used signed Grünwald–Letnikov coefficients in a return recurrence and a reflected nearest-neighbor sector Laplacian. Isolating the current value in a GL derivative does not yield that recurrence's lag and sign convention. A fractional derivative also does not inherit the ordinary logarithmic chain rule. The present work therefore starts from (4)–(10), rather than invoking a fractional Black–Scholes equation.

The retained idea is memory plus diffusion. The changes are explicit: positive fractional-integral weights act on squared standardized innovations; a symmetric estimated graph replaces the artificial sector lattice; positive feedback controls second moments; and risk forecasts are compared chronologically against strong variance references. Hurst identification, convergence of a pricing PDE, and directional price-prediction superiority are not claimed.

D Observed and predicted paths for every ETF

The following figures show every ETF in the two recorded panels, using the 21-session horizon and the full evaluation period. Splitting the sector list across two pages only improves readability; assets are neither selected nor ranked by forecasting performance. The horizontal coordinate is the target window end, and the vertical coordinate is given by (24). Each asset has its own vertical scale, starting at zero. No peak is clipped, and no extra temporal smoothing is applied.

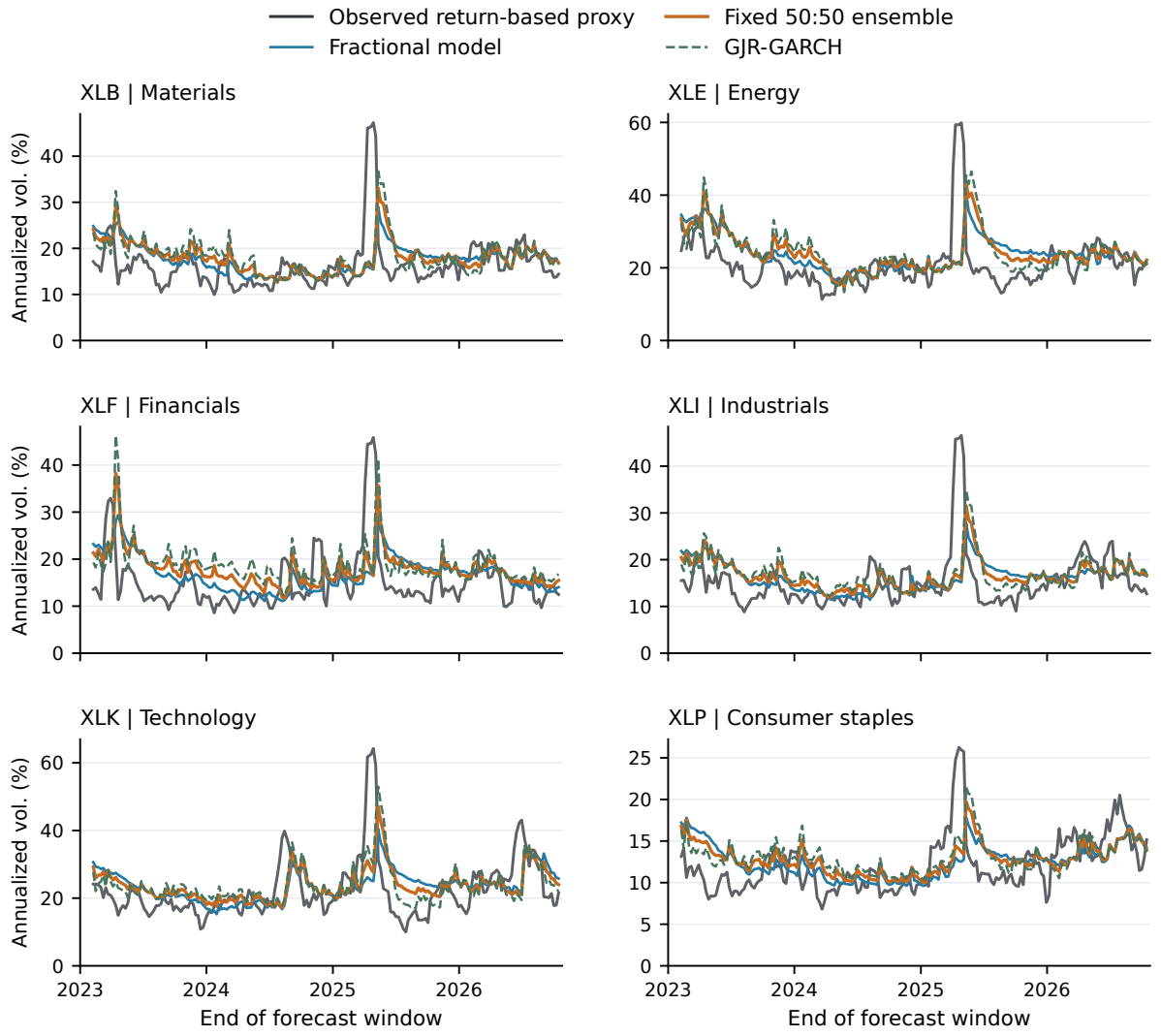


Figure 8: Twenty-one-session observed proxies and forecasts for materials, energy, financials, industrials, technology and consumer staples. Curves use the same methods and colors as Figure 6.

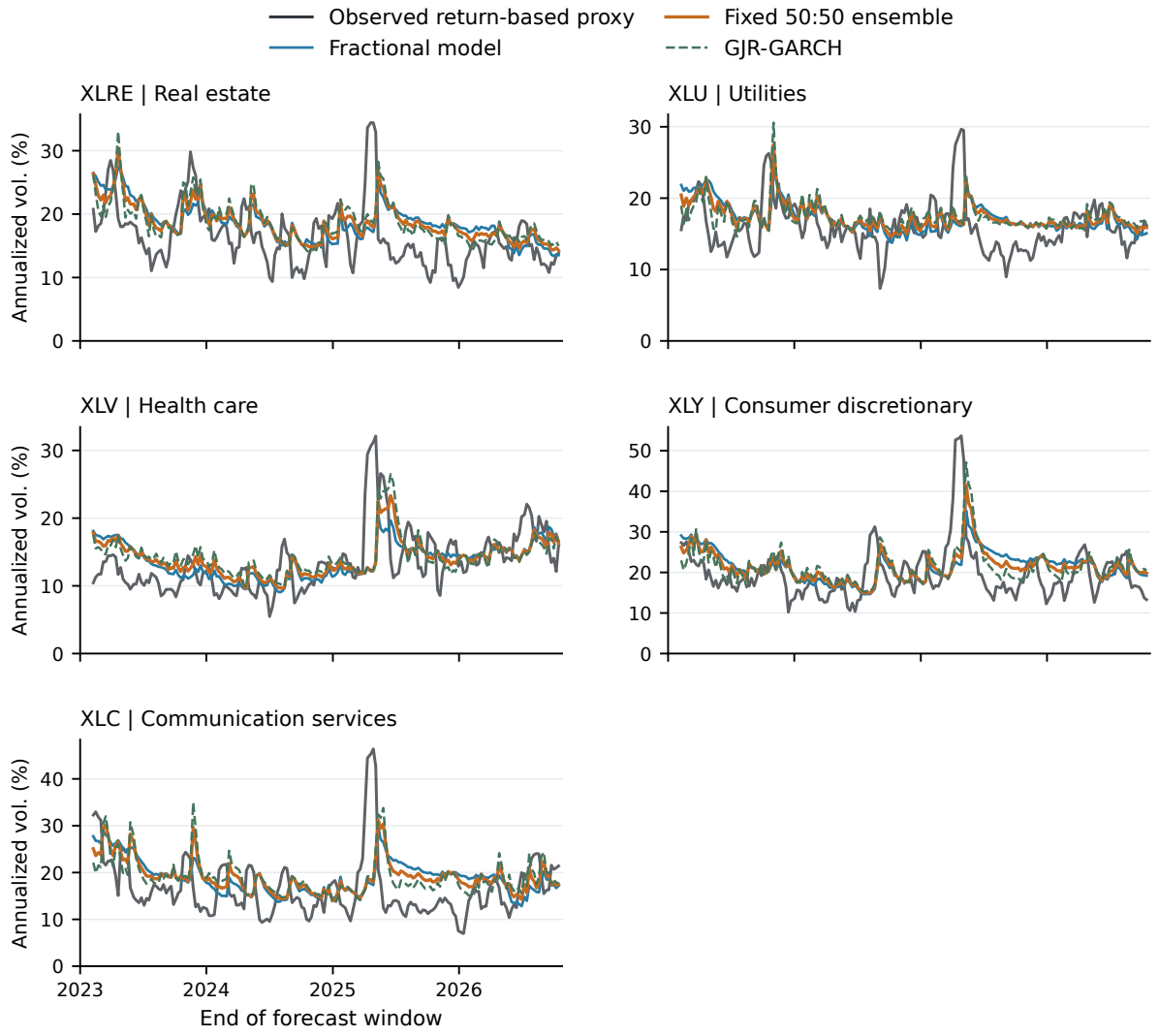


Figure 9: The remaining five sector ETFs: real estate, utilities, health care, consumer discretionary and communication services. Every sector is included exactly once across these two figures.

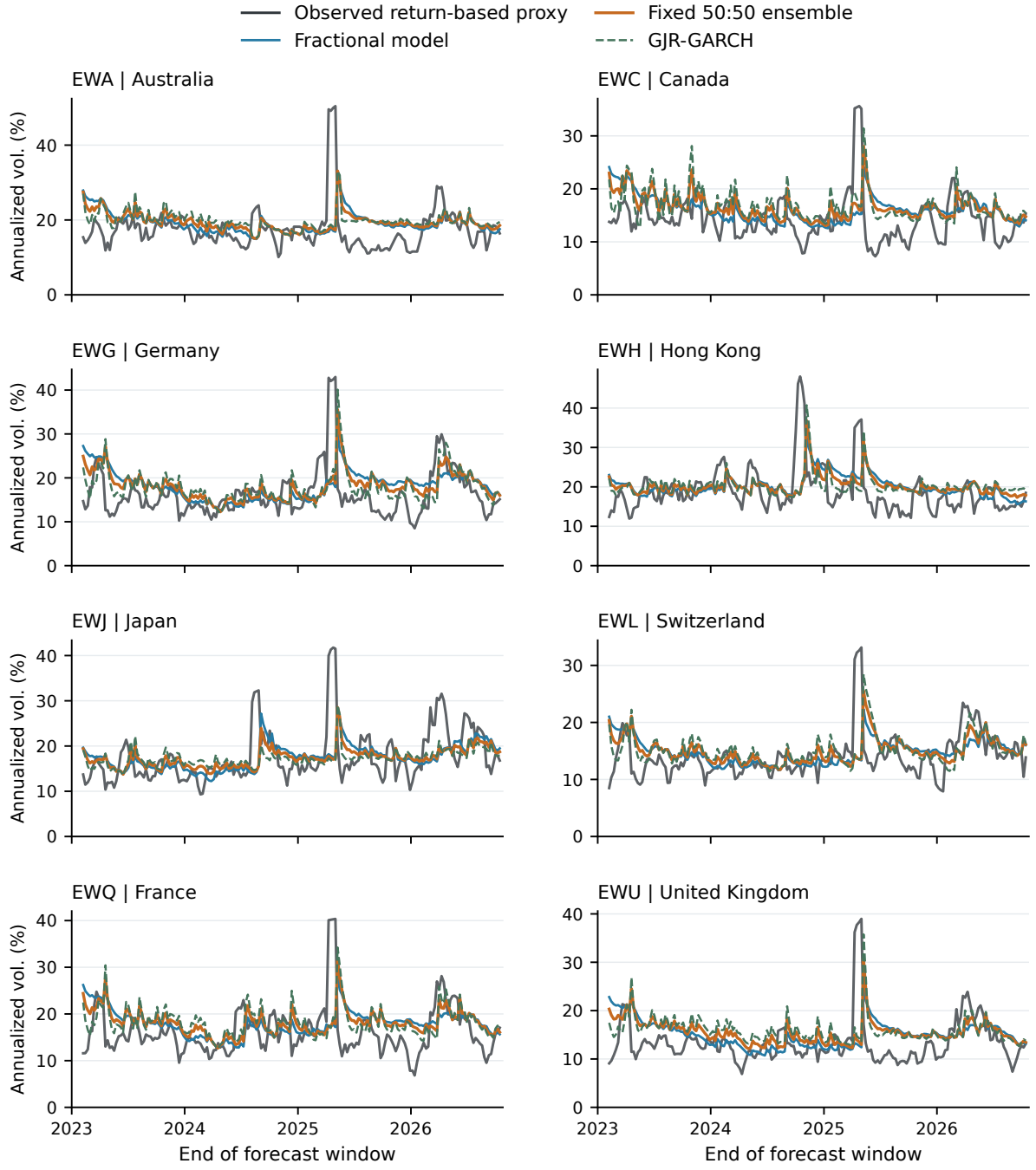


Figure 10: Twenty-one-session observed proxies and forecasts for all eight pre-specified country ETFs. The observed series is a noisy daily-return-based proxy; the plots do not imply prediction of the underlying ETF price direction.

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